

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Pooja B (192411091) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Artificial Intelligence

CDI-AT3

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1. Types of Intelligent Agents

a) Simple Reflex Agents

Acts only on the current input using predefined rules. It does not remember previous events.

Example: Turns ON the light when motion is detected.

b) Model-Based Agent

Maintains an internal model (memory) of the environment. Makes decision based on both current and past information.

Example: Keeps tracks of previous temperature readings and adjust accordingly.

c) Goal-Based Agent

Chooses actions that help achieve a specific goal. ~~Evaluates~~ different paths before making decisions.

Example: Locks all doors before bed-time to achieve home security.

d) Utility Based Agent

Selects the actions that provides the highest overall benefit. Considers multiple factors like comfort, energy consumption, and security.

Example: Adjusts AC and lighting to maximize comfort while minimizing electricity usage.

Functions of Each Agent in the Smart Home

1. Simple Reflex Agent

Controls lights based on motion detection.

2. Model Based Agent

Learns occupancy patterns and remembers previous states.

3. Goal-Based Agent

Maintains desired temperature and security goals.

4. Utility Based Agent

Balances comfort, energy efficiency, and security for the best overall performance.

Best Agent for the Smart Home

A Hybrid Agent (Model-Based + Goal-based
+ Utility-Based)

Reasons:

- * Learns user performance and their preferences over time.
- * Achieves goals like maintaining comfort and security.
- * Optimizes electricity usage.
- * Makes intelligent decisions by considering multiple factors simultaneously.

Conclusion:

A Hybrid Intelligent Agent provides the best performance because it combines learning, planning, and optimization, making the smart home more comfortable, energy efficient and secure.

2.

Robot Navigation

1. Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the lowest path cost first.

How it helps:

- * Finds the least-cost path to the exit.
- * Suitable when different paths have different movement costs.

2. Iterative Deepening Search (IDS)

IDS repeatedly performs Depth-First Search with increasing depth limits until the goal is found.

How it helps:

- * Explores the maze level by level
- * Requires very little memory
- * Guarantees finding the solution if one exists.
- * ~~Suitable~~ when the depth is unknown.

Feature	UCS	IDS
Time complexity	$O(b^{1+(c*/2)})$	$O(b^d)$
Space complexity	High	Low
Advantages	Finds optimal path	Uses less memory and is complete
Disadvantages	High memory usage	Repeats node expansion

where

b = branching factor

d = depth of solution

c^* = optimal path cost

s_1 = minimum step cost

Intelligent Agents in Maze Navigation

Simple Reflex Agent: Moves based only on current perception; may get stuck in loops.

Model-Based Agent: Remembers visited locations and avoids revisiting them.

Goal-Based Agent: Plans actions to reach the maze exit.

Utility Based Agent: Selects the safest and least-cost path.

Effects of Agent Design on Decision Making

- * Better agent design leads to smarter decisions.

- * Memory helps avoid repeated paths.

- * Goal-oriented planning improves navigation.

- * Utility evaluation selects the most efficient and safest route.

Best Combination

Agent Type: Model Based.

Goal Based.

Search Strategy: UCS

Justification

- * The model based component remembers explored paths.
- * The goal based component focuses on reaching the exit
- * UCS guarantees the optimal (lowest-cost) path.
- * This combination provides efficient, reliable, and intelligent maze navigation.

Conclusion:

A model based Goal based Agent with UCS is the most effective solution for navigating an unknown maze, as it remembers explored paths and finds the optimal route to the exit.